

Assignment 3: GST & Pricing in the Indian Motorcycle Market (2020-25)

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1 Executive Summary

India's two-wheeler market is one of the fastest-growing vehicle segments globally, with pricing and tax policy playing a central role in consumer accessibility. This report analyses bike prices and Goods and Services Tax (GST) across segments in India from 2020 to 2025 using a publicly available dataset of Indian bike models. Key findings reveal significant price variation across segments, with the premium category recording the highest average prices and GST contributions over the study period. This report recommends targeted GST relief for commuter-segment bikes to improve affordability for lower-income riders.

2 Introduction

India is the world’s largest two-wheeler market by volume, with millions of units sold annually, making it a critical segment of the broader automotive industry (**bikes2025?**). The two-wheeler segment serves as the primary mode of transport for middle and lower-income households across both urban and rural areas of India. Since the introduction of the Goods and Services Tax (GST) in 2017, tax rates on two-wheelers have varied by engine capacity, directly influencing retail pricing and consumer purchasing decisions. The period from 2020 to 2025 saw significant market disruption due to the COVID-19 pandemic, global supply chain shortages, and a gradual shift toward electric vehicles. These factors collectively drove price increases across most bike segments, making affordability a growing concern for everyday commuters. This report investigates how ex-showroom prices and GST contributions have evolved across different bike segments in India over this five-year period. The dataset used, sourced from Kaggle, contains pricing and tax data for a range of Indian bike models spanning 2020 to 2025 (**bikes2025?**). The analysis focuses on segment-wise price trends, GST burden distribution, and year-on-year price comparisons. Understanding these dynamics is relevant to consumers, manufacturers, and policymakers seeking to balance industry growth with consumer affordability. The following sections outline the methodology, present key findings, and provide recommendations based on the analysis.

3 Methodology

This analysis uses a dataset of 1,000 Indian motorcycles spanning eight major brands across three market segments from 2020 to 2025. The dataset contains 18 variables including engine displacement (cc), mileage (kmpl), factory price, GST rate, GST amount, ex-showroom price, on-road price, and buyer behaviour indicators.

Descriptive statistics were computed to summarise key metrics by market segment. Pricing variables were analysed to measure the GST burden across segments. Buyer behaviour categories and price sensitivity levels were used as proxies for consumer response to pricing changes. All analysis was conducted in R using the tidyverse and kableExtra packages. As shown in Table 1, the three market segments differ substantially in price and GST burden.

Table 1: Summary of key metrics by motorcycle market segment

Segment	No. of Models	Avg Engine (cc)	Avg Mileage (kmpl)	Avg On-Road (INR)	Avg GST (INR)
budget	258	112.9	63.0	97,444	18,032
mid	407	181.0	42.3	180,279	33,912
premium	335	376.4	27.9	315,860	62,207

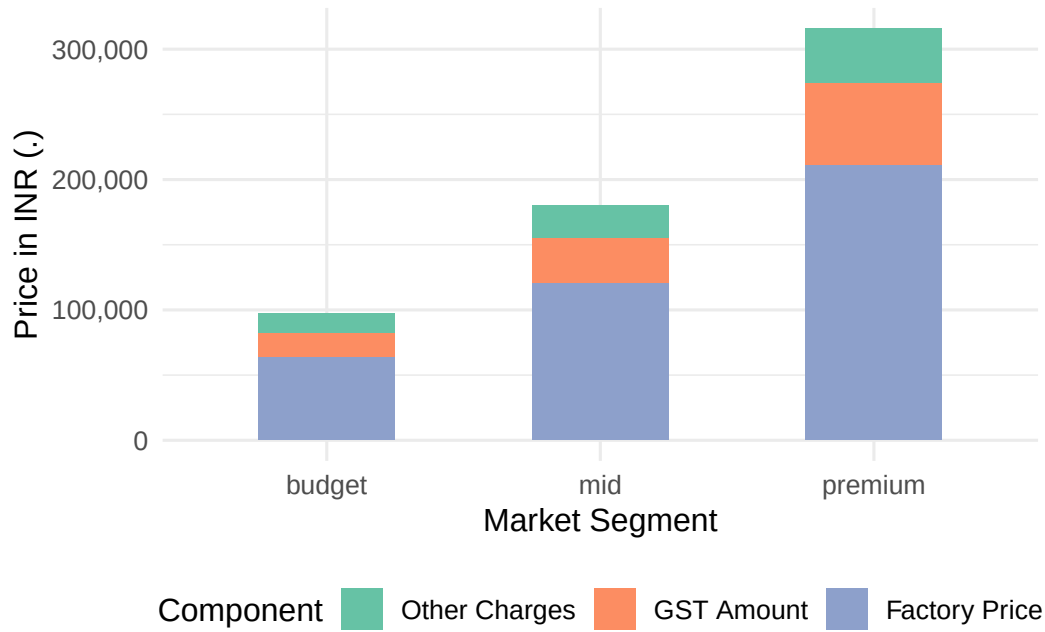


Figure 1: Average price component breakdown by market segment

4 Results

As shown in Table 1, the three market segments differ substantially in engine size, mileage, premium models feature larger engines averaging 376.4 but deliver a lower average mileage of 27.9 kmpl.

The financial data highlights how taxation impacts final pricing. As illustrated in Figure 1, the total cost scales across segments. The average Goods and Services Tax (GST) amount rises from INR 18,032 for budget motorcycles to INR 62,207 for premium motorcycles. This tax burden directly raises the average on-road price from INR 97,444 in the budget segment to INR 315,860 in the premium segment.

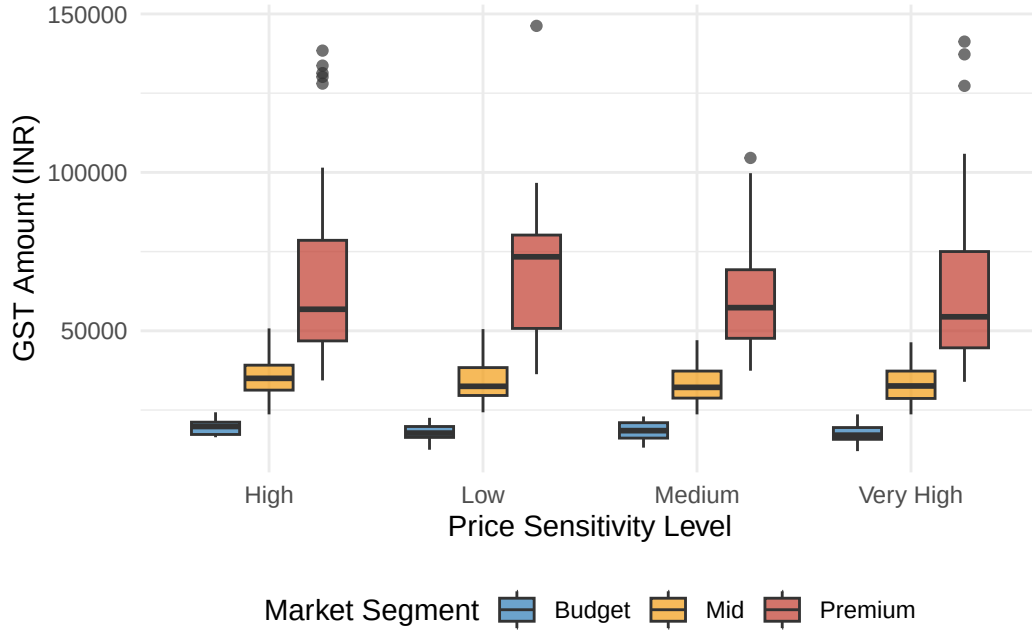


Figure 2: Distribution of GST amount paid by price sensitivity level and market segment.

Figure 2 shows that the GST burden within each market segment remains consistent across all four price sensitivity levels. Regardless of whether a consumer reports low, medium, high, or very high price sensitivity, the absolute GST amount they pay is determined entirely by their market segment rather than their sensitivity level. Premium-segment buyers pay approximately 50,000 to 80,000 INR in GST across all sensitivity groups, while budget-segment buyers consistently pay 15,000 to 20,000 INR. This confirms that the structural impact of the uniform 28% GST rate is segment-driven: budget consumers, who are most constrained by price increases, receive no tax relief despite their lower purchasing capacity.

5 Discussion, Conclusion and Recommendations

5.1 Discussion

Table 1 shows that the three market segments differ substantially in both technical specifications and GST burden. Premium models carry larger engines and higher on-road prices, with an average GST amount exceeding three times that of budget-segment motorcycles, despite all segments facing the same 28% statutory rate.

Figure 2 deepens this finding. The GST burden within each segment remains consistent across all four price sensitivity levels, confirming that a consumer's tax burden is determined entirely

by market segment rather than by individual price sensitivity. Budget-segment buyers, despite reporting higher price sensitivity, receive no reduction in their GST obligation. Mukherjee (Mukherjee 2024) demonstrates that India’s GST system places a disproportionate relative burden on lower-expenditure households, a pattern clearly reflected here: budget buyers pay less in absolute rupees but bear a heavier relative cost given their lower purchasing capacity (Ali and Gharpure 2021).

This structural asymmetry directly shapes brand competitiveness. Brands concentrated in the budget segment face consumers who are price-constrained and cannot benefit from the current flat-rate tax structure, limiting those brands’ pricing flexibility. Premium brands, by contrast, operate in a segment where the same proportional tax produces far less behavioural pressure on buyers.

5.2 Conclusion

The uniform 28% GST rate imposes structurally unequal absolute tax burdens across Indian motorcycle market segments. Budget-segment buyers pay the least GST in rupees yet experience the greatest relative pressure, while premium-segment buyers absorb far higher amounts with considerably less behavioural disruption. Figure 2 confirms that this burden is determined by segment alone and does not adjust to consumer purchasing capacity. A single statutory rate therefore produces asymmetric competitive pressure across the market.

5.3 Recommendations

Budget-segment manufacturers should treat any price increase as a last resort and accompany it with tangible improvements in product features or after-sales service so that consumers perceive added value rather than a simple cost transfer. Premium-segment manufacturers can absorb or pass on GST-related cost increases with less risk of volume loss, but should monitor price sensitivity data regularly to detect early shifts in buyer behaviour. Policymakers may consider whether differentiated GST rates across segments could reduce the disproportionate relative burden on budget buyers without significantly disrupting premium market dynamics (Mukherjee 2024).

References

Ali, Amina Mohsin, and Ravindra Gharpure. 2021. “Impact Assessment of Customer Motivation on Market Behaviour of Two-Wheeler Automobile Industry.” *Journal of Management Information and Decision Sciences* 24 (5): 1–10.

Mukherjee, Sacchidananda. 2024. "Is GST Regressive in India? Distribution of Tax Burden Across Consumer Groups." *Economic and Political Weekly* 59 (24). <https://www.epw.in/journal/2024/24/special-articles/gst-regressive-india.html>.